

JomExam

ESAT Mathematics 1 — Sample Paper

Six questions in the style of ESAT Mathematics 1. Allow 9 minutes — the real paper's 90-second pace. No calculator. One option is correct in each question, and there is no negative marking.

Worked solutions with the trap named follow the questions. The full experience — 27 questions, 40 minutes, and a report naming the skill behind every wrong answer — is free at jomexam.com.

Question 1. The price of a jacket is reduced by 25% in a sale, and the sale price is later increased by 25%. The final price is what percentage of the original price?

- A 100%
- B 93.75%
- C 95%
- D 106.25%
- E 98%

Question 2. Positive numbers x and y satisfy $x^2 + y^2 = 34$ and $xy = 15$. What is the value of $x + y$?

- A 7
- B $\sqrt{34}$
- C 8
- D 15
- E 64

Question 3. Each interior angle of a regular polygon is 150° . How many sides does the polygon have?

- A 10
- B 15
- C 9
- D 12
- E 30

Question 4. A fair four-sided spinner (numbered 1 to 4) and a fair six-sided spinner (numbered 1 to 6) are spun together. What is the probability that the two scores add to exactly 5?

- A $\frac{1}{5}$
- B $\frac{1}{4}$
- C $\frac{1}{10}$
- D $\frac{5}{24}$
- E $\frac{1}{6}$

Question 5. Two positive whole numbers have highest common factor 6 and lowest common multiple 180. One of the numbers is 30. What is the other?

- A 36
- B 150
- C 60
- D 6
- E 72

Question 6. A sequence begins 2, 6, 12, 20, 30, ... with the pattern continuing in the same way. What is the tenth term?

- A 90
- B 100
- C 132
- D 110
- E 120

Worked solutions

Question 1 — answer B.

1. Reducing by 25% multiplies by 0.75; increasing by 25% multiplies by 1.25.
2. $0.75 \times 1.25 = 0.9375$.

The trap: The trap is 100% — the instinct that the changes cancel. They act on different bases: the increase applies to the smaller sale price, so it recovers less than the cut removed.

Question 2 — answer C.

1. $(x + y)^2 = x^2 + y^2 + 2xy = 34 + 30 = 64$.
2. x and y are positive, so $x + y = 8$.

The trap: Solving for x and y individually is a quadratic you do not have time for. The identity does it in one line — the question is testing whether you reach for $(x + y)^2$ before you reach for x .

Question 3 — answer D.

1. Each exterior angle is $180^\circ - 150^\circ = 30^\circ$.
2. Exterior angles sum to 360° : $360/30 = 12$ sides.

The trap: Working from the interior-angle formula gets there too, but slowly and with algebra to slip on. The exterior angle is the two-line route.

Question 4 — answer E.

1. List the pairs summing to 5: (1,4), (2,3), (3,2), (4,1) — four outcomes.
2. Total outcomes: $4 \times 6 = 24$. Probability $4/24 = 1/6$.

The trap: The trap is counting outcomes as if the spinners were the same: (1,4) and (4,1) are different outcomes. Count ordered pairs over 24, always.

Question 5 — answer A.

1. For any two numbers, $\text{HCF} \times \text{LCM} = \text{product of the numbers}$.
2. Other number = $6 \times 180 / 30 = 36$.

The trap: Without the identity this is a search through factor lists against the clock. $\text{HCF} \times \text{LCM} = \text{product}$ is the single most time-saving fact in the number questions — and 150 is waiting for anyone improvising.

Question 6 — answer D.

1. The terms are $n(n + 1)$: 1×2 , 2×3 , 3×4 , 4×5 , 5×6 ...
2. Tenth term: $10 \times 11 = 110$.

The trap: The differences tell you it is quadratic, but chasing the formula through differences is slow. Spotting the product structure $n(n + 1)$ answers it instantly.